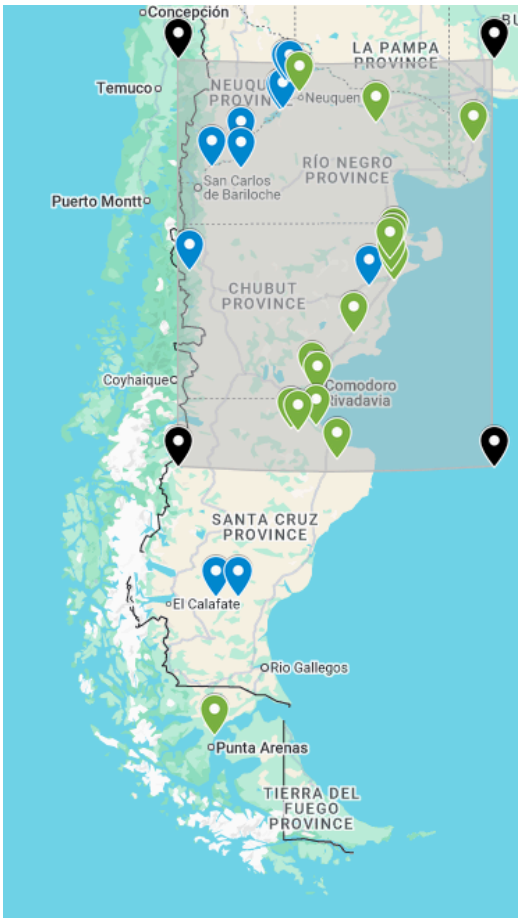
Can compound wind-hydro energy droughts in Patagonia be predicted from large scale climate modes, and if so, how far in advance?

Locations and Geography

Hydro

El Chocón Dam
Piedra del Águila Dam
Alicurá Dam
Pichi Picún Leufú Dam
Arroyito Dam
Futaleufú Dam
Florentino Ameghino Dam
Loma de la Lata dam
Planicie Banderita and El Chañar
Under Construction:
Néstor Kirchner Dam
Represa Jorge Cepernic



Wind

Rawson Wind Farm
Parque Eolico Trelew
Parque Eolico Chubut Norte
Puerto Madryn Wind Farm
Genpat Wind Farm
Manantiales Behr Wind Farm
Antonio Morán Wind Farm
Cañadón León Wind Farm
Bicentenario Wind Farm
Los Hércules Wind Farm
Koluel Kayke Wind Farm
Pomona Wind Farm
Patagónicos Wind Farm
Vientos Neuquinos
Garayalde Wind Farm
Villalonga / Vientos Bonaerenses

Conclusion found that a bounding box of -47.5, -72.0, -38.0, -62.0, as while there are 2 dams (Néstor Kirchner Dam and Represa Jorge Cepernic) and a wind farm further south, it would decrease accuracy for minimal gain (both dams are still under construction and Parque Eólico Vientos Patagónicos (Patagónicos Wind Farm) is a small local area wind farm of 10MW, and so not worth the data dilution to include.

Project Definition

Goals

1. Define a compound renewable energy drought index and catalogued historical events.
2. Quantify the strength and lag of teleconnections between climate modes and my index.
3. Build a forecast model and compare it against meaningful baselines.

Answer

I need to be able to say either *"Yes, SAM/ENSO provides X months of lead time with Y skill"* or *"No, these droughts are not predictable from large scale modes, and here's why",* or a partial result in between (it may predict well for wind but not hydro or vise versa).

What is out of scope?

- Actual generation data from wind farms or hydroelectric plants
- Grid integration or economic analysis
- Sub monthly timescales
- Factoring in Climate Change, while it will include future predictions, it will not factor in global warming

Variables and Datasets

Variable	ERA5 Name	Raw Units	Converted Units	Conversion/Note
100m U-wind	100m_u_component_of_wind	m/s	m/s	100m is selected for turbine height
100m V-wind	100m_v_component_of_wind	m/s	m/s	wind_speed = $\sqrt{(u^2 + v^2)}$
Total precipitation	total_precipitation	m/day	mm/month	Multiply by 1000 and days in month

Runoff	runoff	m/day	mm/month	Same conversion as precip
2m temperature	2m_temperature	K	ºC	Subtract 273.15
snowmelt	snowmelt	m/day	mm/month	Multiply by 1000. Andean pixels only will be relevant

Spatial Domain

- **Bounding box:** 38°S–47.5°S, 72°W–62°W
- **Resolution:** 0.25° monthly
- **Processing:** Area weighted average across all land grid cells within bounds for each variable

Temporal Domain

- **Range:** 1979–2025
- **Frequency**: Monthly

The reason for the selection of the 1979-2025 is that while ERA5 is available from 1940 the climate indices (especially SAM) are most reliable post 1978. Starting in 1979 gives 47 years, long enough for robust climatology and trend analysis while keeping all data sources aligned.

Climate Indices

Index	Full Name	Source	URL	Format	Temporal Coverage
SAM	Southern Annular Mode (Marshall)	British Antarctic Survey	Available from BAS website	Monthly values	1957–present
ONI	Oceanic Niño Index	NOAA CPC	Available from NOAA CPC	Monthly, 3 month running mean	1950–present
IOD/DMI	Indian Ocean Dipole Mode Index	NOAA PSL or BOM	Available from NOAA or BOM	Monthly SST difference	1870–present

Derived Variables

Derived Variable	Computation

Wind speed	$\sqrt{(u^2 + v^2)}$ from u and v components
Standardised wind anomaly	Remove monthly climatology, divide by monthly standard deviation
Standardised runoff anomaly	Same approach
Compound Wind-Hydro Drought Index (WHDI)	Mean of standardised wind and runoff anomalies
Rolling WHDI variants	3 month and 6 month rolling averages of WHDI

Data Quality Checks

- No gaps in monthly timeseries
- Reasonable value ranges: wind speed positive, precipitation not negative, temperature within expected bounds for Patagonia (~-15°C to +35°C)
- Seasonal cycles make physical sense, wind should peak in a specific season, precipitation should show expected pattern. If they don't, something went wrong in download or processing
- Climate indices align temporally, all index time series cover the full 1979–2025 period with no gaps
- Area averaging is land-only, I apply a landsea mask so I am not averaging in ocean grid cells, particularly on the Atlantic coast side of the bbox where the boundary matters

Methodology

Standardisation

For each variable (wind speed, runoff), for each calendar month:

1. Compute the 1979–2025 climatological mean for that month
2. Compute the standard deviation for that month
3. Standardised anomaly = (value - mean) / standard deviation

I create an energy drought index with the equation: Wind Hydro Drought Index (WHDI)

$$WHDI = \frac{standardised_wind + standardised_runoff}{2}$$

While the option could be to weigh each variable depending on their importance, I decided to just assume equal weighting as I would be entering into an infrastructure analysis if I start to weight based upon “importance” of wind vs hydro. Despite this, it could be tested changing the weighting of both variables.

Index	Window	What It Demonstrates
WHDI1	1 month	Instantaneous compound anomaly
WHDI3	3 month rolling	Short term compound drought
WHDI6	6 month rolling	Medium term compound drought

WHDI Value	Category
Below -1.0	Moderate energy drought
Below -1.5	Severe energy drought
Below -2.0	Extreme energy drought

Teleconnection Analysis

Lag correlation will function for each climate variable, I correlate lag 1 to 12 with WHDI. I also correlate with the individual components to see how the climate mode affects each component (more or less for one or the other).

As monthly climate data here is autocorrelated, I have fewer data points than initially suggested. As such I use effective degrees of freedom where r1 is the lag 1 of autocorrelation of the time series:

$$n_{eff} = n \times \frac{1-r_1}{1+r_1}$$

Composite Analysis

1.
For all climate modes positive months are where index > 0.75 standard deviations and negative months are where index < 0.75 standard deviations.
2.
For every ERA5 variable, I calculate the mean anomaly during the positive and negative climate mode months, and create a spatial map of the difference between them.
3.
For each grid cell, I test whether the composite mean is significantly different from zero using a t test. I mask insignificant areas on the maps.

It will be necessary to do this for each season individually, as climate modes influences vary by season.

ML Forecasting Setup

Sub question

"*Which forecasting paradigm is most appropriate for climate teleconnection prediction: regularised linear methods, tree based ensembles, boosted ensembles, or sequence models?"*

Target Variable

- Primary: WHDI3 at time t+3 (3 month lead forecast)
- Secondary: WHDI at time t+1 (1 month lead for comparison)

Feature Set

Group A:
- SAM at times t, t-1, t-2, t-3
- ONI at times t, t-1, t-2, t-3
- IOD at times t, t-1, t-2, t-3
- Seasonal Indicators

Group B:
- WHDI3 at times t, t-1, t-2
- Standardised wind anomaly at t
- Standardised runoff anomaly at t
- Snowmelt anomaly at t

Models

Model	Family	Structural Assumption	What It Tests
ElasticNet	Regularised Linear	Teleconnection relationships are linear, with both sparsity (L1) and grouped feature retention (L2) for correlated climate indices	Is the climate mode signal simple enough that a linear model captures it? If ElasticNet performs comparably to complex models, the underlying relationship is straightforward
Random Forest	Tree Ensemble	Non-linear feature interactions matter, but each prediction is independent — no temporal structure assumed	Do non linear interactions between climate modes (e.g., SAM × ENSO interaction effects) improve prediction beyond linear combination?
XGBoost	Boosted Ensemble	Sequential residual correction on tree predictions, with	Does iterative error correction improve on Random Forest's independent tree approach for

		regularisation to prevent overfitting on small samples	this specific prediction problem?
LSTM	Recurrent Neural Network	The temporal ordering of input sequences carries information beyond what handcrafted lag features capture	Does explicitly modelling the sequence of climate states add value, or do lagged features already capture the relevant temporal memory?

Each model represents a different assumption about how climate information maps to drought outcomes.

- If ElasticNet wins or ties the teleconnection is approximately linear, and complex methods are unnecessary for operational use
- If tree based methods win the non linear interactions between modes matter
- If LSTM wins, temporal dynamics contain information that static lag features miss
- If all models perform similarly the predictive signal is robust and model choice is secondary to feature choice
- If all models fail the signal isn't there

LSTM

Input:

- A sequence of the most recent 6 monthly timesteps, where each timestep contains all Group A and Group B features for that month

Initial Architecture:

- Single LSTM Layer (32-64 units)
- Dropout Layer (0.2-0.3)
- Dense output layer (single value, WHDI3 prediction)

Baselines

Baseline	Definition	What It Tests
Climatology	Predict zero (the long term mean, since data is standardised)	Can any model beat "nothing unusual will happen"?
Persistence	Predict WHDI3(t+3) = WHDI3(t)	Can any model beat "things stay the same"?
Local only model	Best performing model using only Group B features	Do climate modes add skill beyond knowing current local conditions?
Climate only model	Best performing model using only Group A features	Can large scale modes alone predict local compound droughts without any local information?

Evaluation Strategy

I use a walk forward cross validation with 8 folds:
Fold 1: Train 1979-2000, Test 2001-2003
Fold 2: Train 1979-2003, Test 2004-2006
Fold 3: Train 1979-2006, Test 2007-2009
Fold 4: Train 1979-2009, Test 2010-2012
Fold 5: Train 1979-2012, Test 2013-2015
Fold 6: Train 1979-2015, Test 2016-2018
Fold 7: Train 1979-2018, Test 2019-2021
Fold 8: Train 1979-2021, Test 2022-2025

For each fold, I evaluate all 4 models plus all 4 baselines, for the metrics:

- RMSE
- MAE
- R^2
- Skill Scores for each baseline $1 - \frac{rmse_model}{rmse_baseline}$

Aggregate reporting:

- Mean and standard deviation of each metric across folds
- Performance trajectory plot showing skill per fold over time
- Statistical significance of model differences (paired test across folds)

The main comparison points are going to be, for all effective models (where skill is more effective than baseline):

- Do the models agree on which features matter the most?
- Which group of variables dominate?
- Do climate modes change importance depending on lag?

Anomaly Detection

Fit Isolation Forest on climate mode features
Catalogue anomalous months
Cross reference with WHDI events and forecast errors

Notebook Plan

Notebook 1: Data Collection

Input:

- ERA5 CDS API credentials
- Climate index URLs (BAS for SAM, NOAA CPC for ONI, NOAA/BOM for IOD)
- Bounding box coordinates: 38°S–47.5°S, 72°W–62°W

Process:

1. Download ERA5 monthly data for 1979–2025:
    - 100m u-component of wind
    - 100m v-component of wind
    - Total precipitation
    - Runoff
    - 2m temperature
    - Snowmelt
2. Compute wind speed from u/v components: $\sqrt{(u^2 + v^2)}$
3. Convert units:
    - Precipitation: m/day to mm/month
    - Runoff: m/day to mm/month
    - Snowmelt: m/day to mm/month
    - Temperature: K to °C
4. Apply land sea mask to exclude ocean grid cells
5. Compute area weighted averages across study region for each variable
6. Download SAM, ONI, IOD monthly timeseries
7. Align all time series to common date index (1979-01 to 2025-XX)
8. Merge into single combined dataset

Quality Checks:

- No gaps in any timeseries
- All values within physically reasonable ranges
- Seasonal cycles visible in raw plots (sanity check)
- Climate indices temporally complete across study period

Figures:

- Map of study region with bounding box and infrastructure locations
- Raw time series plots for each variable (visual sanity check)

Notebook 2: Climatology & Index Construction

Input:

- combined_dataset.csv from Notebook 1

Process:

1. Compute monthly climatology for all variables
2. Compute seasonal cycles and plot
3. Compute long term trends for wind speed, runoff, precipitation, temperature
4. Standardise wind speed and runoff by calendar month
5. Construct WHDI: mean of standardised wind and standardised runoff
6. Compute rolling WHDI1, WHDI3, WHDI6
7. Catalogue drought events from WHDI3:
    - Event starts when WHDI3 drops below -1.0
    - Event ends when WHDI3 returns above -1.0
    - Record start date, end date, duration, peak severity for each event
8. Compute drought frequency statistics
9. Examine the individual components: are wind droughts and runoff droughts occurring simultaneously or independently?

Figures:

- Seasonal cycle plots for wind speed and runoff
- Long term trend plots with regression lines
- WHDI3 timeseries with drought events highlighted
- Scatter plot of standardised wind vs standardised runoff (shows whether compound events are common or rare)
- Drought frequency comparison between early and late periods

Notebook 3: Teleconnection Analysis

Input:

- whdi_timeseries.csv from Notebook 2
- climate_indices.csv from Notebook 1
- ERA5 NetCDF files (for spatial composite maps)

Process:

1. **Lag correlation analysis:**
    - Correlate SAM with WHDI3 at lags 0–12 months
    - Correlate SAM with standardised wind and standardised runoff separately
    - Repeat for ONI and IOD
    - Compute effective degrees of freedom for significance testing
    - Plot correlation matrices showing lag structure
2. **Seasonal stratification:**
    - Repeat lag correlations for all seasons separately
    - Identify which seasons show strongest teleconnection signal
    - Determine whether wind and hydro components respond to the same mode in the same season
3. **Composite analysis (spatial):**
    - Define positive/negative SAM composites (top/bottom quartile months)
    - For each ERA5 variable, compute spatial mean anomaly during positive vs negative SAM

    - Map the difference with significance stippling
    - Repeat for El Niño vs La Niña composites
    - These maps show where within the study region the teleconnection is strongest
4. **SAM trend analysis:**
    - Document the observed SAM trend over 1979–2025

Figures:

- Lag correlation bar plots for each climate mode x each target variable
- Seasonal stratification heatmap (mode x season x component)
- Composite anomaly maps for wind speed (positive vs negative SAM)
- Composite anomaly maps for precipitation/runoff (positive vs negative SAM)
- Same for ENSO composites
- SAM trend timeseries

Notebook 4: ML Forecasting & Anomaly Detection

Input:

- combined_dataset.csv from Notebook 1
- whdi_timeseries.csv from Notebook 2
- Key lags identified in Notebook 3

Process:

Part A: Feature Engineering

1. Construct feature matrix using lags informed by Notebook 3 findings
2. Split into Group A (climate modes) and Group B (local conditions)
3. Create combined feature set (Group A + B)
4. For LSTM: reshape into sequences of 6 monthly timesteps
5. Handle any edge effects from lagging (first few months will have missing lag values)

Part B: Model Training & Evaluation

6. Implement walk forward CV (8 folds)
7. For each fold:
    - Train all four models on training period
    - Compute all four baselines
    - Evaluate on test period
    - Record all metrics
8. Run three feature configurations through best performing model:
    - Group A only (climate modes)
    - Group B only (local conditions)
    - Group A + B combined
9. Compare to determine whether climate modes add skill

Part C: Feature Importance

10. Extract ElasticNet coefficients
11. Compute permutation importance for Random Forest
12. Extract XGBoost gain based importance
13. SHAP values for XGBoost if time permits
14. Compare across models, do they agree on what matters?

Part D: Anomaly Detection

15. Fit Isolation Forest on climate mode feature space
16. Catalogue anomalous climate months
17. Cross reference with WHDI drought events
18. Cross reference with forecast errors from Part B
19. Write interpretation paragraph

Figures:

- Model comparison box plot (RMSE across folds for each model)
- Skill score comparison against each baseline
- Performance trajectory across folds
- Feature importance comparison across interpretable models
- Group A vs Group B vs combined skill comparison
- Observed vs predicted time series for best model
- Anomalous months marked on WHDI timeseries

Notebook 5: Synthesis & Visualization

Input:

- All results CSVs from Notebooks 2-4
- ERA5 NetCDF files (for spatial visualization)
- Drought catalogue from Notebook 2
- Composite anomaly fields from Notebook 3
- Model performance results from Notebook 4
- Infrastructure location coordinates

Process:

Part A: Interactive Climate Mode Map

1. Load composite anomaly fields from Notebook 3 (spatial grids for positive/negative SAM, El Niño/La Niña)
2. Build interactive map with toggleable layers:
    - Base layer: study region with topography
    - Infrastructure overlay: wind farm and hydro plant locations with capacity labels

    - Selectable climate state: positive SAM, negative SAM, El Niño, La Niña
    - Display layer: wind speed anomaly and/or runoff anomaly spatial pattern for selected state
3. Add colour scale showing anomaly magnitude
4. Add significance masking (grey out or reduce opacity for non-significant grid cells)

Part B: Summary Dashboard

5. Compile headline findings:
    - Number of compound drought events identified
    - Longest and most severe compound drought
    - Drought frequency trend (early vs late period)
    - Strongest teleconnection (which mode, which lag, which season)
    - Best performing model and its skill vs baselines
    - Whether climate modes add skill beyond local conditions
6. Build dashboard layout with:
    - Key numbers panel at top
    - WHDI3 time series with drought events highlighted
    - Model comparison summary
    - Teleconnection strength summary graphic

Part C: Figures

7. Identify the 6 to 8 most important figures from Notebooks 2-4
8. Remake with consistent styling
9. Export as high resolution PNGs for Conclusion

Part D: Results Compilation

10. Build single compiled results table containing all key numbers
11. Write narrative summary answering the research question in one paragraph
12. Document all caveats and limitations in one place

Figures:

- Interactive climate mode map
- Dashboard layout with key findings
- 6-8 polished versions of the most important analytical figures
- Study region overview map with infrastructure